

2023.

Q4. (a).

$$H(z) = \frac{Y(z)}{X(z)} = \frac{(1 - e^{j\omega_0} z^{-1})(1 - e^{-j\omega_0} z^{-1})}{(1 - r e^{j\omega_0} z^{-1})(1 - r e^{-j\omega_0} z^{-1})}$$

$$= \frac{1 - z^{-1} \cos \omega_0 + z^{-2}}{1 - r z^{-1} \cos \omega_0 + r^2 z^{-2}}$$

$$\Rightarrow X(n) - \cos \omega_0 \cdot X(n-1) + X(n-2) = Y(n) - r \cos \omega_0 Y(n-1) + r^2 Y(n-2)$$

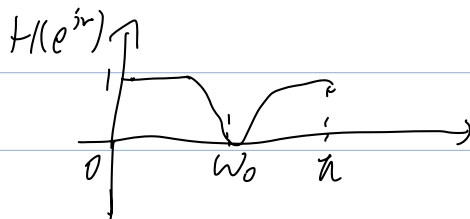


(b). $r < 1$

(c). Two zeros : $z = e^{\pm j\omega_0}$

Two poles : $z = r e^{\pm j\omega_0}$

* when r is small



* when r is big



cd) $f_s = 32 \text{ KHz}$

$f_c = 8 \text{ KHz}$

its digital angular frequency is:

$$\omega_c = 2\pi \frac{f_c}{f_s} = \frac{\pi}{2}$$

In the bilinear transform, we have

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_c}{2} = 2f_s \tan \frac{\omega_c}{2} = 64 \text{ k rad/s}$$

or $f_c = \frac{\Omega_c}{2\pi} = 10.186 \text{ kHz}$

with this frequency, we can design a butterworth filter.

Q1. Fourier Transform.

and by using bilinear transform.
we can

ca).

python code:

$x = s.astype(float) / 2^{15}$

convert
to floating point.

$X = \text{np.fft.fft}(x)$ FFT

signed 16 bit

$K = N // 2$

$X_{\text{-pos}} = X[0:K+1]$

only one side.

$\text{mag} = \text{np.abs}(X_{\text{-pos}}) / N$ sometime $2/N$

$\text{eps} = 1e-12$

$X_{\text{dB}} = 20 * \text{np.log10}(\text{mag} + \text{eps})$

to dB

$\text{freq} = \text{np.arange}(0, K+1) * f_s / N$ x-axis

$\text{plt.plot}(\text{freq}, \text{mag})$

plt.show

plot.

(b). $f_s = 48 \text{ kHz}$ $T_s = 101$

$N = 480,000$

$\Delta f: \frac{f_s}{N} = 0.1 \text{ Hz}.$

What we need to remove. $f < f_c = 100 \text{ Hz}.$

pos $k_{c, \max} = \frac{f_c}{\Delta f} = 1000$

neg $k_{c, \min} = N - \frac{f_c}{\Delta f} = 479,000$

$Y[k] = \begin{cases} 0, & k = 0, \dots, 1000 \text{ and } k = 479,000, \dots, 480,000. \\ X[k], & \text{otherwise} \end{cases}$

python code:

$X = \text{np.fft.fft}(x)$

$\Delta f = f_s / N$

$k_{\max} = \text{int}(100 / \Delta f)$

$X[0:k_{\max}+1] = 0$

$X[-k_{\max}:] = 0$

$y = \text{np.fft.ifft}(X).real.$

(c). $80 \sim 300 \text{ Hz}$ bass $\rightarrow$ boost

$300 \sim 4000 \text{ Hz}$ remain

$\begin{cases} < 50 \text{ Hz} \\ > 8000 \text{ Hz} \end{cases} \rightarrow \text{attenuate.}$

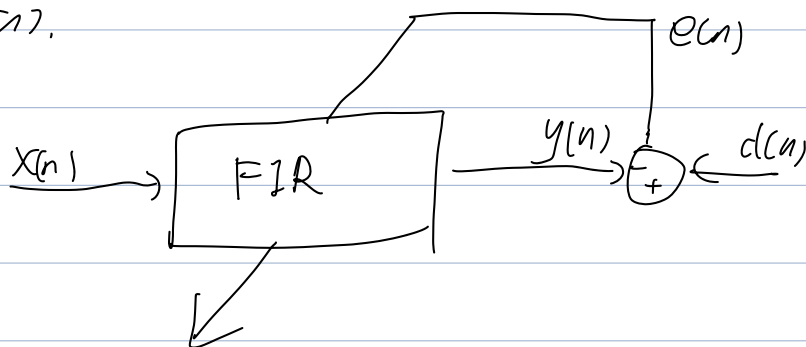
Qd1. Fridge hum is a stationary narrowband tone. (50Hz)
can be removed by narrow notches without affecting ...

Chatter is speech-like, broadband and highly overlapping in freq. with wanted speaker.

also time varying.

Q3. Adaptive FIR and LMS.

Con.



To remove f_c 50Hz interference, we first need to define some key signal:

$x(n)$ = noise signal (50Hz)

$d(n)$ = ECG signal (with noise)

$e(n) = d(n) - x(n)$ = output signal.

$$\Delta h_m = -\eta \cdot \frac{\partial \frac{1}{2} e(n)^2}{\partial h_m} = \eta e(n) x(n-m)$$

In the process of updating FIR parameter h_m , the $e(n)$ is approaching to 0.

However $e[n]$ will not exactly be 0,

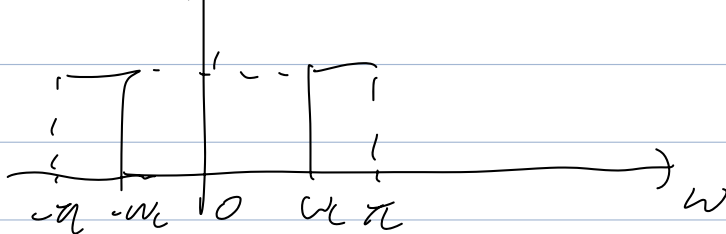
∴ it will contain any signal component that is not correlated with $x[n]$ (noise)

∴ the output signal $e[n]$ is cleaned up signal.

[b], pros :

cons : CMS needs a reference signal and parameter tuning.

Q2.
(a). $|H(\omega)|$



when $n \neq 0$.

$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{-\omega_c} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{\omega_c}^{\pi} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \cdot \frac{1}{jn} \left(e^{jn(-\omega_c)} - e^{jn(-\pi)} \right)$$

$$\begin{aligned}
& + \frac{1}{2\pi} \cdot \frac{1}{jn} (e^{jn\pi} - e^{jn\omega_c}) \\
& = \frac{1}{2\pi jn} (e^{jn\omega_c} - e^{-jn\omega_c}) \\
& = \frac{1}{2\pi jn} \cdot 2j \cdot \sin(n\omega_c) \\
& = \frac{1}{\pi n} \sin(n\omega_c)
\end{aligned}$$

$$\begin{aligned}
\text{when } n=0: \quad h(n) &= \frac{1}{2\pi} (-\omega_c + \pi + \pi - \omega_c) \\
&= 1 - \frac{\omega_c}{\pi}
\end{aligned}$$

$$\text{Therefore } h(n) = \begin{cases} 1 - \frac{\omega_c}{\pi}, & n=0 \\ \frac{1}{\pi n} \sin(n\omega_c), & n \neq 0. \end{cases}$$

(b),